

No.	Solution	Remarks
4(a)	A progressive wave is one in which energy is transferred from one point to another without the transfer of matter or medium. A transverse wave is one in which the oscillation of the particles is perpendicular to the direction of energy transfer. (Note: a clear reference to direction of energy transfer is required.)	[1] [1]
4(b)	$\Delta\phi = \frac{\Delta t}{T} \times 2\pi$ $= \frac{1.4}{4.0} \times 2\pi$ $= 2.199 \text{ rad}$ $\approx 2.2 \text{ rad (2 s.f.)}$	[1] for correct substitution [1] for correct answer (before rounding off)
4(c)	Intensity of wave I, amplitude of wave A and radius r. $I \propto A^2$ $I \propto \frac{1}{r} \quad (\text{for 2D circular wavefront})$ $\therefore A^2 \propto \frac{1}{r}$ $r_Q A_Q^2 = r_P A_P^2$ $r_Q = r_P \left(\frac{A_P^2}{A_Q^2} \right)$ $= 150 \left[\frac{(2.0)^2}{(1.0)^2} \right]$ $= 600 \text{ mm}$	[1] [1] for correct working
4(d)	$\Delta\phi = \frac{\Delta x}{\lambda} \times 2\pi$ $2.199 = \frac{600 - 150}{\lambda} \times 2\pi$ $\lambda = 1286 \text{ mm}$ $\approx 1300 \text{ mm (2 s.f.)}$	[1] for correct working [1] for correct answer

5

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